Ashutosh Sudhir Kumbhar

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EDUCATION

Master of Software Engineering **August 2024 - May 2026**
Arizona State University Tempe, AZ, USA

PROFESSIONAL EXPERIENCE

Software Engineer Intern - Synopsys **May 2025 - May 2026**

- Architected scalable HPC simulation job scheduling pipelines for semiconductor design workflows on AWS EKS using Terraform and Kubernetes, integrating RabbitMQ and KEDA for event-driven job scheduling across 15 Linux nodes processing 30K+ daily simulation workloads for clients including Apple, TSMC, NVIDIA, Intel and other leading semiconductor companies.
- Debugged, monitored, and optimized production Kubernetes clusters using Datadog observability, reducing queue latency and deployment time while improving job throughput.
- Cut client cloud costs by 30% and provisioning time by 80% by building HPCDS, a self-service HPC provisioning service in Nest.js that let clients spin up, scale, and tear down their own infrastructure on demand, managing Terraform state through the Spacelift GraphQL API.
- Secured Kubernetes clusters with 15+ Kyverno policies, blocking privileged pods and untrusted images and fixing 40+ vulnerabilities.

Software Engineer - Western Union Financial Services **January 2023 - August 2024**

- Reduced request latency from ~30 seconds to 400ms and setup time from 15 minutes to under 1 minute by building a proxy tool that linked local frontends to hosted QA/UAT services, boosting frontend developer productivity.
- Built WUPOS Database-Compare, a Tauri (Rust) desktop tool that automated configuration diffs between production and QA/UAT, cutting config mismatch checks from ~2 hours to under 5 minutes and helping developers replicate production issues 25% faster.
- Integrated a Generative AI (LLM) summarization layer presenting configuration drift with probable root causes, accelerating root-cause analysis across 10+ Scrum teams.
- Reduced production incidents by 15% by resolving 40+ critical bugs across services handling millions of financial transactions and adding **150+** Playwright end-to-end regression tests, raising test coverage from 40% to 80%.
- Migrated **5+** core modules of a legacy monolith to distributed Java Spring Boot microservices and Angular 18 (TypeScript) using Test Driven Development (TDD), enabling independent deployments across **10+** Scrum teams.

Full-Stack Software Engineer - MindsClik Digital Solutions **January 2022 - January 2023**

- Architected a white-label eCommerce platform in Flutter with a shared component library (product pages, cart, checkout, payments), cutting new store launches from **~10 weeks to 4 weeks** and earning a 4.8/5 Google Play Store rating.
- Optimized data fetching with GraphQL and PostgreSQL query indexing, cutting average API response times from ~800ms to 300ms.
- Automated provisioning of isolated AWS environments for **20+ client stores** using Terraform modules for EC2, S3, and VPC networking, cutting environment setup from ~2 days to 30 minutes.

PROJECT EXPERIENCE

Code-Studio - Kubernetes-Based Lab-as-a-Service and Online Compiler: https://hub.docker.com/r/8766834856/compiler

- Designed and deployed a Kubernetes-based compute platform on Linux, providing isolated, on-demand dev environments for 200+ students.
- Developed a browser-based IDE using Vue.js with TypeScript enabled, Node.js, and node-pty, supporting interactive Linux terminals, 10+ programming languages, and seamless integration with learning management systems such as Moodle and Canvas.
- Automated the provisioning, monitoring, and cleanup of containerized coding sessions while enforcing CPU, memory, and execution-time limits for workload isolation.
- Diagnosed Linux, container networking, and Kubernetes scheduling failures using logs and pod events, improving system reliability.

Prospective - Distributed Cold Outreach GTM Pipeline: https://prospective.akumbhar.com

- Built a distributed outreach pipeline with independent prospecting, ranking, and mailer services on Kubernetes, using RabbitMQ to decouple services and KEDA to autoscale mailer pods on queue depth, achieving a 35% reply rate across ~300 emails/day sent via 10 mailboxes.
- Designed an adapter layer over Apollo and Clay RESTful APIs using Go Lang with idempotent writes and retry with backoff, storing ranked prospects in Postgres and enforcing per-account send limits that maintained >80% deliverability with zero domain blacklists.
- Integrated an LLM for prospect scoring and personalized email drafting with structured outputs and a validation step, plus a reply classifier that feeds outcomes back into ranking, continuously improving targeting precision.

Restaurant Point of Sale Application (POS): https://tasteandtwist.co.uk/

- Engineered Angular NX powered Mono-repo POS app, streamlined restaurant workflows via Firebase Cloud Functions and Firestore.
- Devised a client-facing kiosk application mirroring **McDonald's** self-service functionality, reducing staff workload by **30%**.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, Go Programming Language, Rust, JavaScript, TypeScript, SQL
Distributed Systems & Design: Concurrency, Multi-threading, OOP Design Patterns, Event-Driven Architectures, Scalable Systems
Frontend & Mobile Frameworks: Angular, Vue.js, Electron.js, Flutter
Cloud & Infrastructure: AWS, Docker, Kubernetes, CI/CD, Linux, RabbitMQ, Apache Kafka, Terraform, Datadog, Prometheus, ArgoCD
Database & Operations: MongoDB, PostgreSQL, Redis, ETL Framework, Job Schedulers, Agile, Git, DevOps, AI-Assisted Development